LONGEST UNIQUE SUBSTRING

Aim

To write a Java program to find the longest substring in a given string that contains no repeated characters.

Description

The program reads a string from the user. It checks substrings starting at each character and adds characters until a repeated character is found. Whenever a longer valid substring is found, it is stored as the result. Finally, the program displays the longest unique substring and its length.

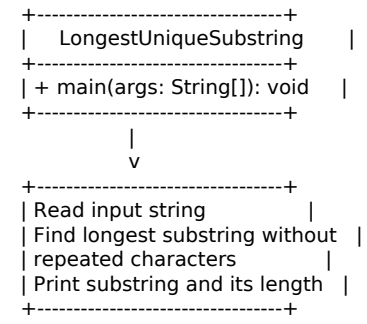
Software Requirements

Hardware: Computer or laptop. Software: Java Development Kit (JDK) and a Java IDE or text editor such as Notepad, VS Code, or Eclipse.

Algorithm

1. Start the program.
2. Create a Scanner object and read a string from the user.
3. Initialize the result string as empty.
4. For each starting position i, initialize an empty temporary string.
5. Move through the string from position i using j.
6. If the current character already exists in the temporary string, stop this inner loop.
7. Otherwise, add the character to the temporary string.
8. If the temporary string is longer than the current result, update result.
9. Display the longest unique substring and its length.
10. Close the Scanner and stop.

UML Diagram



Code

```
import java.util.Scanner;

public class LongestUniqueSubstring
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();
        String result = "";

        for (int i = 0; i < str.length(); i++)
        {
            String temp = "";

            for (int j = i; j < str.length(); j++)
            {
                char ch = str.charAt(j);

                if (temp.contains(String.valueOf(ch)))
                    break;

                temp = temp + ch;
            }

            if (temp.length() > result.length())
                result = temp;
        }

        System.out.println("Longest unique substring: " + result);
        System.out.println("Substring length: " + result.length());

        sc.close();
    }
}
```

Output Screenshot

```
Enter a string: abcabcbb
Longest unique substring: abc
Substring length: 3
```

Note: The output above is an example run using the input abcabcbb.

Result

Thus, the Java program to find the longest substring without repeated characters was written and executed successfully.